

Money Market Disconnect

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A repurchase agreement (repo) is a source of cash and collateral. We document that the money market is more segmented when the collateral motive prevails. Two crucial aspects of the central bank framework lead to this disconnect: banks' access to the central bank's deposit facility and assets' eligibility for quantitative easing (QE). We show that repo rates lent by banks with access to the deposit facility and secured by QE eligible assets are more collateral-driven and disconnected from funding-based money market rates. Our results are relevant for different monetary policies and have suggestive implications for the monetary policy pass-through. (*JEL* E40, E43, E50, E52, E58, G18)

Received November 30, 2021; editorial decision February 4, 2023 by Editor Ralph Kojien. Authors have furnished an Internet Appendix, which is available on the Oxford University Press Web site next to the link to the final published paper online.

The money market is crucial for the efficient allocation of funding and financial securities as well as for preserving financial stability. Furthermore, “a deep and integrated money market is a precondition for an efficient monetary

We appreciate helpful comments from Ralph Kojien (editor) and three anonymous referees. We also thank Farshid Abdi, Andrea Barbon, Roman Baumann, Alexander Bechtel, Patrick Bolton, Martin Brown, Pierre Collin-Dufresne (discussant), Stefano Corradin, Cody Couture (discussant), Sebastian Infante (discussant), Florian Heider, Marie Hoerova, Sebnem Kalemli-Özcan, Christian Keuschnigg, Oliver Krek (discussant), Thomas Nellen, Patrick Schaffner, Paul Söderlind, Olivier Wang (discussant), and Kerstin Westergren (discussant); conference participants at the American Economic Association (2021), Young Swiss Economist Meeting (2021), ECB-RFS Macro-Finance Conference (2021), Eastern Finance Association (2021), SFI Research Days (2021), Swiss Society of Economics and Statistics Congress (2021), World Finance Conference (2021), 52nd Conference of the Money, Macro and Finance Society (2021), SNB Research Conference (2021), 16th Central Bank Conference on the Microstructure of Financial Markets (2021), RCEA Conference (2022), and ECB Money Market Conference (2022); and seminar participants at the University of St. Gallen (2020), Goethe University Frankfurt (2020), Norges Bank (2021), and Swiss National Bank (2021) for their valuable comments and suggestions. This work is supported by the Swiss National Science Foundation [grant number P1SGP1_188111]. An earlier version of this paper circulated under the title “Monetary Policy Disconnect.” Supplementary data can be found on *The Review of Financial Studies* web site. Send correspondence to Hannah Winterberg, hannah.winterberg@unisg.ch.

The Review of Financial Studies 36 (2023) 4158–4189

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<https://doi.org/10.1093/rfs/hhad022>

Advance Access publication April 3, 2023

policy, since it ensures an even distribution of central bank liquidity and a homogeneous level of short-term interest rates” (European Central Bank 2011). Given its essential role in the financial and economic system, we want to understand frictions in the money market and *how* they lead to market segmentation, which can create rate dispersion and can be detrimental to market quality.

The repo is the main money market instrument. It is a secured short-term loan that serves a dual role to obtain cash and collateral. A particular government bond (“special repo”) or any bond from a predefined basket (“general collateral or GC repo”) can be used as collateral.¹ We focus on the European repo market, which is a centrally cleared money market that eliminates frictions of bilateral, over-the-counter (OTC) markets and thus provides the ideal laboratory to examine the dual nature of repos as the underlying mechanism leading to market segmentation. Banks can operate either as *lenders* of cash in repos to invest liquidity safely at a market-based return (funding motive) or to source the collateral (collateral motive). The idea underlying this paper is that the money market becomes more segmented when repos are predominantly collateral-driven. Empirically, a weaker correlation between repo rates and funding-based money market rates is a sign of this segmentation. To document this mechanism, we investigate two crucial aspects of the central bank’s institutional framework as a source of this segmentation.

The first source is that only some banks have access to the ECB’s deposit facility. In an environment where GC rates drop *below* the central bank’s deposit facility rate (DFR) (henceforth $GC < DFR$), access banks have the outside option to store liquidity at the central bank instead of engaging in funding-based repo trades; at the same time, access banks continue to have a motive to trade for collateral-purposes in the repo market. In such an environment, we expect an endogenous selection of access banks into more collateral-driven repo segments that are less connected to funding-based money market rates. At first, it seems surprising that secured lending rates of access banks are more disconnected from other money market benchmark rates. At the same time, our proposed mechanism is intuitive as the deposit facility is the outside lending option which is only available for access banks. Banks without access to the deposit facility, by contrast, do not have the option of accessing the deposit facility and, therefore, select themselves into more funding-driven repo segments which behave more like the unsecured market.

In line with this intuition, we show that access banks lend more in collateral-driven repo segments in the $GC < DFR$ environment and as a result, their repo lending rates are more disconnected from the EONIA as the funding-based money market benchmark rate. In such an environment, an increase in the EONIA rate by one percentage point translates into an increase in lending

¹ In the United States, a special repo is sometimes also referred to as a “specific” repo.

rates involving access banks of only 28 basis points (bps; pointing to market segmentation) compared to an almost one-to-one reaction of banks without access (pointing to high integration).

The second source of our mechanism relates to the eligibility criteria of the QE program which specify that only a given set of assets can be purchased by the ECB. For this part, we focus on the “special” repo segment. QE purchases lead to asset scarcity and repo rates secured by those assets become more special (Arrata et al. 2020; Corradin and Maddaloni 2020). As a consequence, we expect repos with QE eligible assets as collateral to become predominantly driven by collateral demand disconnecting them from funding-based money market segments.

In line with this idea, we find that in the period after the introduction of QE, an increase in the EONIA rate by one percentage point is associated with a 50% lower sensitivity of QE eligible assets. We also apply the initial QE implementation provisions retrospectively to compare time trends between (hypothetically) eligible and noneligible assets, which creates a difference-in-differences estimation setting. In addition to the diverging patterns during QE, we observe a similar behavior of (hypothetically) eligible and noneligible assets in the periods prior to QE, suggesting a causal impact of central bank asset purchases and resultant asset scarcity on money market segmentation.

Our results are relevant for different monetary policies. We study the introduction of the ECB’s two-tier system for remunerating excess reserve holdings in 2019 (“tiering”), which is a natural experiment for our mechanism. By exempting part of the excess reserve holdings held at the ECB from negative rates, it created an even stronger incentive for access banks to store funds at the deposit facility. As a result, we show that the lending rates of access banks are more disconnected from the EONIA rate after the introduction of the tiering system. Widening deposit facility access to foreign banks as in Switzerland (Kraenzlin and Nellen 2015) is a similar illustration of this mechanism. Amendments to the Capital Requirements Regulation (CRR), at the heart of which is the (temporary) exclusion of central bank reserves from the calculation of the leverage ratio, could also manifest money market segmentation as it encourages banks (with access) to store additional funds at the ECB’s deposit facility. Our results are also informative for the recent discussion about new short-term benchmark rates as they underline that measuring the price of cash becomes increasingly difficult in a collateral-driven repo market environment.

Since the EONIA rate is the operational target rate of the ECB, our results about money market segmentation also have *suggestive* implications for the monetary policy transmission process. The idea is that a “wider dispersion in short-term money market rates” could cause “a reduction in the efficacy and transmission of monetary policy” (Bank for International Settlements 2017, p. 32). To test this, we compute a dispersion indicator of repo rates based on Duffie and Krishnamurthy (2016) by taking into account access status and QE eligibility. The results show that interest rates relevant for the

real economy, such as nonfinancial corporate borrowing and housing rates, respond less to changes in the ECB's operational target rate when the dispersion in money market rates is wider. Although not causal, the lower conditional correlation could be relevant for assessing the effectiveness of the monetary policy transmission on a continuous basis.

Our analysis mainly contributes to two strands of the literature. First, we add to the literature on short-term money markets. The innovations we bring are twofold: First, we document the underlying mechanism behind money market segmentation related to repo trades being predominantly driven by collateral demand. And second, we identify two sources of this disconnecting mechanism, that is, access to a central bank's deposit facility and asset eligibility for QE. Arrata, Nguyen, Rahmouni-Rousseau, and Vari (2020) and Corradin and Maddaloni (2020) investigate the effects of QE purchases on the level of special repo rates.² Other papers analyze the unsecured money market, for instance, Kraenzlin and Nellen (2015) examine domestic and foreign banks' behavior in the Swiss franc unsecured money market and Bech and Klee (2011) evaluate the impact of bargaining power in a segmented and unsecured U.S. money market.

Second, we add to the literature on monetary policy, central bank frameworks, and the transmission process (e.g., Duffie and Krishnamurthy 2016; Drechsler, Savov, and Schnabl 2017; Eisenschmidt, Ma, and Zhang 2020). We are the first to shed light on how two key features of the central bank framework, that is, access to central bank facilities and QE purchases, lead to money market segmentation. Our results are relevant for different monetary policies, such as the two-tier system, and the monetary policy transmission.³

1. European Money Market

The focus of this paper is to provide new insights about money market segmentation, its underlying mechanism, and potential implications for monetary policy. For this, we focus on the European money market, which provides a unique setting to analyze those questions.

Figure 1 gives a market overview. It depicts the aggregate trading volumes in the secured repo market and in the unsecured EONIA money market. Trading in the European money market has moved toward the secured segment since the Global Financial Crisis. With a size of more than EUR 8.5 trillion in

² Related is Roh (2021), who points out that the ECB's decision to lend out bonds that it purchased also affects asset scarcity. In addition, the inactive lending of bonds by nonbanks also adds to the scarcity of those assets (Maddaloni and Roh 2021).

³ The existing literature on the effectiveness of monetary policy has mostly focused on its pass-through in the context of specific events, such as announcements or policy changes. Duffie and Krishnamurthy (2016) analyze the effects of the introduction of the reverse repurchase facility and new Basel regulation on the monetary policy pass-through in the United States. Drechsler, Savov, and Schnabl (2017) consider announced changes to the target corridor of the fed funds rate. Eisenschmidt, Ma, and Zhang (2020) analyze the monetary policy transmission in the context of the OTC repo market and dealer market power.

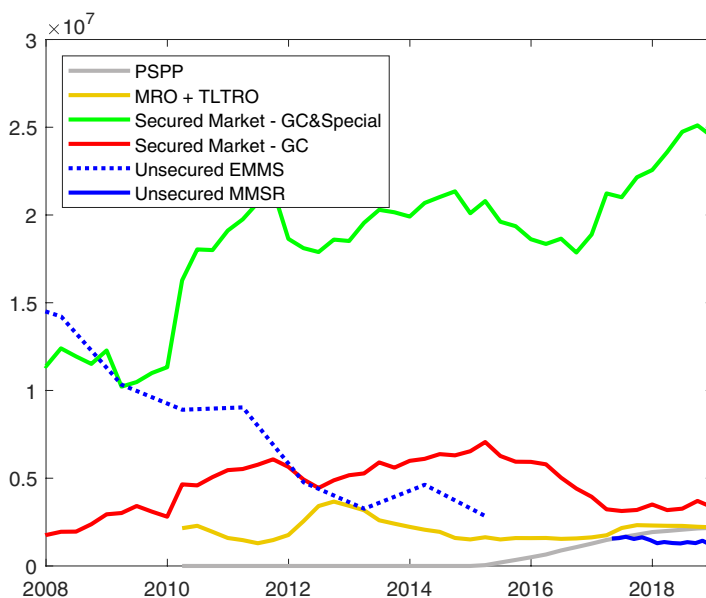


Figure 1
Different market turnovers

The figure depicts the aggregate cumulative quarterly trading volumes in the secured and unsecured market segments as well as the total cumulative QE purchases under the Public Sector Purchase Programme (PSPP) and volumes of the ECB's main refinancing operations (MRO) and targeted longer-term refinancing operations (TLTRO). The data for the secured market refer to the repo market. The data for the unsecured market stem from the Euro Money Market Survey (EMMS) until 2015 and from the Money Market Statistical Reporting (MMSR) thereafter. To be conservative, we sum reported borrowing and lending activity in the unsecured market, which may entail double counting. The data on PSPP purchases and refinancing operations are from the ECB. All data are in euro million.

outstanding contracts (ICMA 2021), the repo market now plays a crucial role for the efficient allocation of money and financial securities. Let's put the size of the repo market into perspective: trading volumes by far exceed volumes of cumulative purchases of the largest ECB QE program, the Public Sector Purchase Programme (PSPP), or of the ECB's main refinancing operations (MRO) and targeted longer-term refinancing operations (TLTRO).

While the repo market has become the main money market segment for banks, the unsecured money market is an important short-term interest rate benchmark. The EONIA rate is the 1-day, unsecured interest rate at which banks lend to one another and represents the funding-based money market benchmark in the euro area. It is "the key ECB interest rate" (European Central Bank 2011) and serves as the operational target rate for the ECB (Cœuré 2018). Similar to the United States, among other countries, the period studied involves a transition in the economy's benchmark interest rate. The ECB has chosen to

replace the EONIA rate with the €STR rate, another unsecured rate, which highlights the ECB's renewed commitment to an unsecured target rate.⁴

The idea developed in this paper is that money market segmentation emerges from the repo market and affects other money market segments, such as the EONIA market. To understand this mechanism, we briefly illustrate the key repo market characteristics.

1.1 Repo market

In a repo trade, two counterparts exchange cash for collateral for a predefined time period with a fixed repurchase obligation. The lender in a repo transaction provides a short-term, typically 1-day, loan (over)collateralized by a sovereign bond and thus benefits from the ability to use the collateral for the time between the purchase and repurchase.⁵ Because of the collateralized nature, repos serve a dual role for cash and collateral. *Lenders* of cash in repos can either look for a safe investment that offers a market-based return or for a way to source the asset serving as collateral.

The European repo market infrastructure is particularly well suited to analyzing money market segmentation as it eliminates many confounding factors. The core segment is the interbank market in which banks trade anonymously via centralized platforms supporting liquidity and a transparent price discovery process. We focus on trades operated through central counterparties (CCP) which constitute the majority of trades in the euro interbank repo market.⁶ CCPs interpose themselves between each lender and borrower which eliminates any concerns regarding individual counterparty risk. The CCP acts as a clearinghouse and applies the same collateral and (credit) risk policies to all CCP members. Our setting, therefore, eliminates frictions, such as bargaining power, counterparty credit risk, and heterogeneous haircuts. The euro area repo market also features a large variety of repos, in both the special and GC segments. The richness of this cross-sectional dispersion allows us to examine the dual role of repos as funding and collateral markets since some repos are more funding-driven while others are more collateral-driven.⁷

⁴ The EONIA rate is determined as the weighted average of the interest rates on unsecured overnight lending transactions denominated in euros, as reported by a panel of contributing banks. In 2017, the ECB announced that the euro short-term rate (€STR) will replace the EONIA as the new short-term interest rate benchmark in the euro area. The €STR rate reflects the wholesale euro unsecured overnight borrowing costs of banks located in the euro area, and thus covers the borrowing cost of a larger set of banks. We consider both rates in our analysis and show that our results are consistent for both rates.

⁵ A repo in the euro area represents a true sale and repurchase of the collateral; that is, the lender becomes the legal owner of the collateral for the repo term.

⁶ CCP-based repos account for more than two-thirds of the total turnover (European Central Bank 2018). The bilateral, OTC repo segment is very small and does not allow for a clear differentiation between general and special collateral repos.

⁷ More detailed information about the European repo market infrastructure can be found in, for example, European Central Bank (2018), Mancini, Ranaldo, and Wrampelmeyer (2016), Nyborg (2016), and Bank for International Settlements (2017).

Figure 2 shows the repo rate development for the average GC rate as well as for the average special repo rates for QE eligible and noneligible assets (illustratively for Germany) relative to the development of the ECB's DFR. Two developments are important: First, GC rates have fallen *below* the DFR at the height of the European sovereign debt crisis in 2012 and during the period of unconventional monetary policy starting in 2015. This is an important regime change for our analysis that we refer to as the $GC < DFR$ environment. It speaks to the importance of access to central bank facilities as it indicates that storing funds at the deposit facility is an attractive alternative for those banks with access to it. It also points to the idea that repo lenders increasingly trade for *collateral* purposes, even in the GC segment, since they are willing to accept a lower remuneration for lending cash in the repo market instead of depositing it at the central bank. Second, during the QE period, repo rates secured by assets eligible for QE have fallen below those for noneligible assets (e.g., Arrata et al. 2020; Corradin and Maddaloni 2020), thus highlighting the role of asset scarcity induced by QE. Market participants are willing to accept a lower interest rate to lend cash against QE eligible than noneligible assets, which points toward *collateral* demand for those eligible assets.⁸

The graphical results suggest that banks with access to the central bank's deposit facility and repos secured by asset eligible for QE behave differently in the money market as those trades are predominantly collateral-driven. Intuitively, we expect repo trades driven by collateral demand to be less connected to funding-based money market segments, thereby giving rise to money market segmentation.

1.2 ECB access

The first source of money market segmentation that we investigate is that only a given set of money market participants are banks that have access to the central bank's refinancing operations and its deposit facility.⁹ In the euro area, the ECB operates two standing facilities: The deposit facility allows for overnight deposits, while the marginal lending facility provides overnight

⁸ The spread between eligible and noneligible rates has been present since the introduction of QE in 2015 and peaked at the end of 2017 when the ECB's Securities Lending Programme was introduced (Brand, Ferrante, and Hubert 2019). For the period prior to QE, we can apply the initial implementation provisions retrospectively to classify assets into (*hypothetically*) eligible and noneligible. The corresponding figure is presented in the Internet Appendix and confirms that the dispersion in levels has increased since the inception of the QE program.

⁹ The importance of access to central bank's facilities has been stressed in the literature. The deposit rate as the rate of remuneration for reserves is a general and important feature of financial intermediaries' decision problems that is incorporated into macro-financial models (Bech and Monnet 2016; Cúrdia and Woodford 2011; Williamson 2019). In these models, a single deposit rate applies uniformly to all market participants and serves as the outside option. Segmentation induced by different access levels to central bank facilities is also supported empirically. For instance, Bech and Klee (2011) argue that the level of the effective federal funds rate was pushed downward by government agencies that could not receive interest on reserves. The analysis in Bech and Klee (2011) is confined to the unsecured market, while we look at segmentation across different secured and unsecured money market segments. In addition, Bech and Klee (2011) attribute their effect to bargaining power, a friction that does not play a role in a centrally cleared setting like ours. Kraenzlin and Nellen (2015) find that banks without access to central bank facilities pay more interest in the unsecured money market to borrow liquidity.

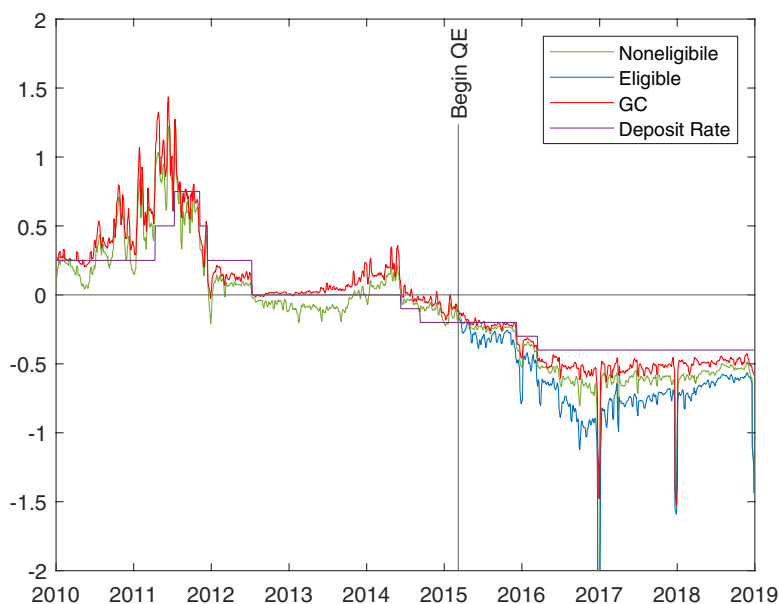


Figure 2
Repo rate development

The figure shows the development of the average volume-weighted GC rate as well as special repo rates for QE eligible and noneligible assets (illustratively for Germany) relative to the development of the ECB's deposit facility rate.

central bank liquidity. Access to the ECB's facilities is, however, limited to eligible counterparties, most importantly to banks that are subject to the Eurosystem's minimum reserve requirements. The minimum reserve system applies to banks and credit institutions established in the euro area. Whether a bank is formed in- or outside of the euro area is unrelated to monetary policy and the repo market, and thus a source of exogenous variation. We exploit the restriction that only euro area banks can access the deposit facility in order to classify counterparties in a repo transaction into *access* and *nonaccess* banks.

To study the different behavior of access and nonaccess banks, we focus on the *lending-side* in GC repos. Banks in the GC repo market can either operate as lenders of cash (a) to have a short-term, safe investment or (b) to source a high-quality collateral. Storing funds at the deposit facility is the outside option for banks with access to it.¹⁰ Figure 3, panel A, depicts, in this context, the development in GC trading volumes for access and nonaccess lenders

¹⁰ Banks could also invest in government bonds directly as opposed to investing liquidity in the repo market or – when having access – placing funds at the deposit facility. However, direct government bond investments have several drawbacks, such as a higher transaction cost, and do not provide the same low-risk and liquid store of value as repos (Bank for International Settlements 2017).

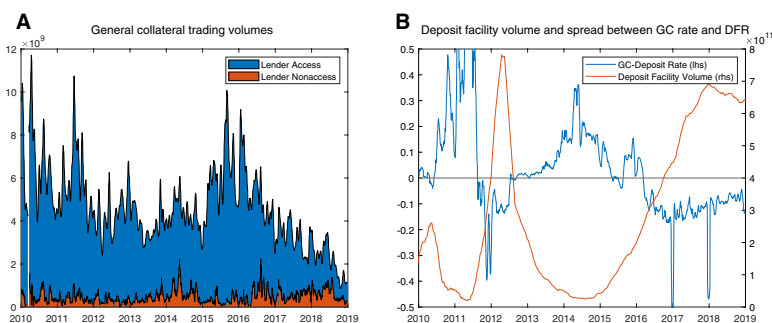


Figure 3

Repo market and deposit facility volumes

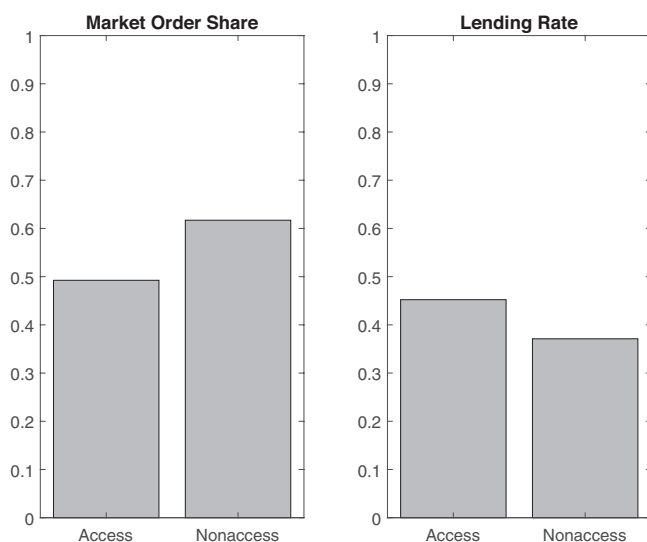
Panel A depicts the total trading volume in the German GC market for trades involving a lender with and without access to the ECB facilities. Panel B depicts the spread between the German GC rate and the ECB's rate on the deposit facility as well as the total volume deposited at the ECB deposit facility.

(illustratively for Germany).¹¹ In the two periods during which GC rates fell below the rate on the deposit facility (i.e., in 2012 and since 2015), we observe a drop in GC trading volume. This drop is accompanied by an increase in the volume of funds deposited at the ECB's deposit facility by access banks (see Figure 3, panel B). For example, since 2015, we observe a drop in GC trading volume to about one-third of its original size. This reduction was mainly driven by banks that had access to the ECB's deposit facility. To our knowledge, this is a new stylized fact suggesting a first form of segmentation between access and nonaccess banks induced by the central bank framework. Access banks increasingly deposit funds at the deposit facility in the $GC < DFR$ environment, whereas nonaccess banks continue to lend in the GC market to obtain a (safe) deposit of liquidity.¹²

The segmentation between access and nonaccess banks relates to the funding-motive in repo trades and comes from the fact that access banks can safely invest liquidity in the repo market or place it at the deposit facility, whereas nonaccess banks can only rely on the former. This implies that nonaccess banks have a higher funding immediacy in repo trades. One illustration of this mechanism is to compare the average market order share and repo rate of access and nonaccess banks for the *same collateral basket* as is shown in Figure 4. Nonaccess lenders have a higher funding immediacy,

¹¹ We present corresponding volume breakdowns by the borrowing side in the Internet Appendix.

¹² The regulatory framework plays a negligible role for our analysis for at least three reasons: First, access and nonaccess banks in our sample are similarly regulated. Second, the new Basel regulation considers all assets under inspection to be of the highest quality (Level 1 assets) from the perspective of the Liquidity Coverage Ratio (LCR) (Bank for International Settlements 2017). Furthermore, all maturities under inspection are shorter than the 30-day LCR cutoff time. Third, the Basel III leverage ratio does not affect the lender in a repo transaction since reverse repos do not enter the Basel III leverage ratio calculation (Ranaldo, Schaffner, and Vasios 2021). In addition, the introduction of the leverage ratio lead to window dressing effects on repo rates, but those were mainly caused by the borrowing counterparties. We provide several robustness checks in which we exclude those trading days.

**Figure 4****Market order share and excess rate frequency**

The left panel depicts the share of market orders and the right panel the excess rate frequency, separately for banks with and without access to the deposit facility. The excess rate frequency refers to how often a bank receives a higher interest rate on their lending activity than the daily average rate in a given basket and is computed according to Bechtel, Ranaldo, and Wrampelmeyer (2023). The figure includes data on all overnight GC repo trades for the time period 2010–2018.

hence, their market order share is higher and the corresponding repo rates are lower.

In the euro area repo market, a variety of *different GC repo baskets* are traded. Separate GC baskets exist for different euro area countries (e.g., German or Italian GC baskets) and for different types of government bonds (e.g., German notes or German long-term bonds). Those GC rates differ and are naturally dispersed. Some of the GC baskets are more collateral-driven (trading at a lower rate) while others are more funding-driven (trading at a higher rate). Figure 5 shows in this context the share of access banks' trading in (a) German government bond baskets (blue line), (b) baskets of core euro area countries (red line), and (c) baskets of peripheral euro area countries including the GC pooling baskets (yellow line).¹³ GC baskets in (a) are more collateral-driven as they ensure that the lender is receiving a high-quality asset, such as a German Bund, as collateral, while GC baskets in (b) and (c) are more funding-driven as they include, for example, the two GC pooling baskets.¹⁴

¹³ The Eurex GC pooling baskets are the main funding baskets in the euro area. They allow for highly rated assets (i.e., investment grade bonds), including assets issued by regional governments or "Pfandbrief" instruments of credit institutions, to be delivered.

¹⁴ We present alternative classifications into collateral-driven and funding-driven trades in the Internet Appendix. The results for all specifications are consistent. The share of nonaccess banks is defined as 100% minus the access banks' share.

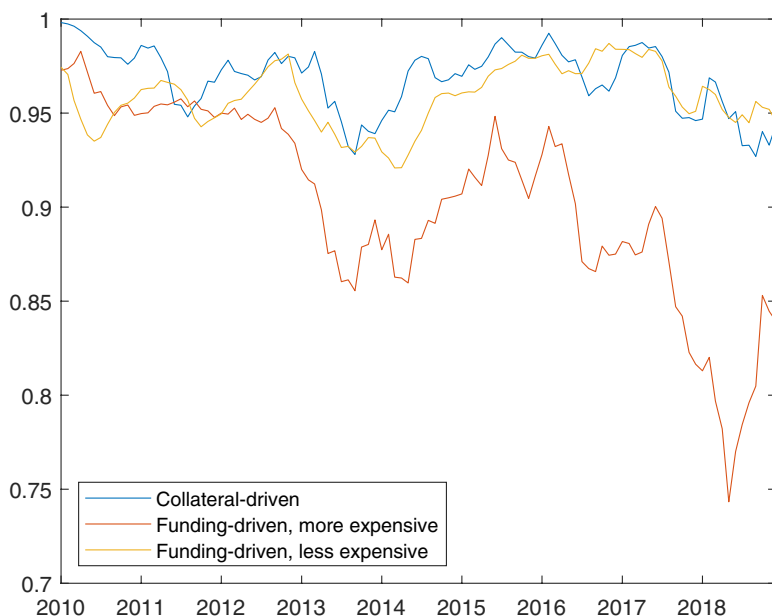


Figure 5
Trading share of access banks

The figure depicts the volume-weighted trading share of access banks in different types of GC baskets. The share is depicted for three subgroups: trades against German government bond baskets (“collateral-driven”), trades involving GC baskets of core euro area countries (“funding-driven, more expensive”), and trades involving collateral baskets of peripheral euro area countries including the GC pooling baskets (“funding-driven, less expensive”).

We observe that in the recent $GC < DFR$ environment, access banks continued to have a collateral motive for lending in the repo market and remained active in collateral-driven baskets; at the same time, they moved out of more expensive, funding-driven baskets trading below the DFR due to their outside option of storing funds at the deposit facility. This means that access banks select themselves into more collateral-driven segments of the repo market in the $GC < DFR$ environment which are less connected to funding-based money market segments. This explains why access banks remain active in funding-driven baskets which yield higher rates than the DFR (those of peripheral euro area countries), while they move out of funding-driven baskets trading at lower rates close to or below the DFR (those of core euro area countries).¹⁵

¹⁵ As some repo rates have been lower than the deposit rate for an extended period, the question arises whether an arbitrage opportunity for access banks exists by borrowing in the repo market and storing the borrowed funds at the deposit facility. This direct comparison of the repo rate and the deposit rate abstracts, however, from the value of the collateral as it assumes that central bank reserves are equivalent to repo loans. As Figure 5 shows, access banks, still trade in the repo market in the $GC < DFR$ environment to obtain collateral. In fact, access banks are willing to accept a lower repo rate compared to the DFR to obtain the collateral, which they would

Changes in the composition of baskets in which access banks trade are inherently connected to the within-basket rate variation. When access banks move out of funding-driven baskets in the $GC < DFR$ environment, the higher funding immediacy of nonaccess banks for those baskets becomes relatively more pronounced, indicating that across and within basket developments go hand in hand.

1.3 Asset eligibility

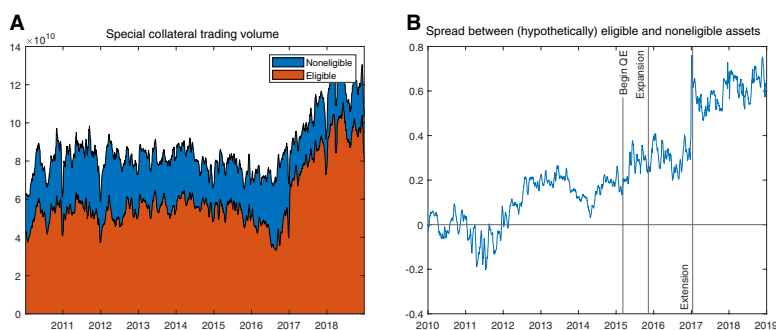
The second source of money market segmentation that we consider are the eligibility criteria of collateral assets for central bank QE purchases.¹⁶ The ECB announced in 2015 its intention to conduct large-scale asset purchases. Since the beginning of these programs, cumulative net purchases amounted to more than 2.5 trillion euro. The PSPP is the largest of the programs implemented in the Eurosystem; it focuses on the purchase of government bonds. The sheer size of these purchases has contributed to scarcity effects for government bonds, which are an important category of safe assets and serve as collateral in repo transactions. QE programs in general aim to influence longer-term rates in an environment where short-term rates are at the zero lower bound (by affecting term premiums; see Eser et al., 2019). An impact of QE-induced asset scarcity on short-term rates is thus an unintended side effect that can increase rate dispersion and lead to market segmentation. The effect of asset purchases on bond scarcity comes on top of tighter regulation of financial institutions under the new Basel framework (e.g., the introduction of the leverage ratio rules). The ECB has, therefore, constituted implementation provisions to limit market impacts and distortions. These provisions specify the assets which the ECB is allowed to purchase.

We exploit the implementation provisions for asset purchases as a source of exogenous variation to classify collateral in a repo transaction into QE *eligible* and *noneligible* depending on the provisions that were valid at a specific point in time.¹⁷ We further apply the initial implementation provisions

forgo by delivering the collateral in a repo trade. Prior research has shown that (repo) arbitrage activity exists but is insufficient to close these “gaps” (Rinaldo, Schaffner, and Tsatsaronis, 2019).

¹⁶ The interplay of central bank asset purchases, financial intermediation, and collateral has been featured prominently in the theoretical literature. Gertler and Karadi (2013) show that if limits to arbitrage exist in the banking sector, central bank purchases of securities cause yields to fall. Araújo, Schommer, and Woodford (2015) stress that the direction of the impact of asset purchases depends on the way collateral constraints are affected. Piquard and Salakhova (2019) highlight how monetary policy affects unsecured and secured markets in a different way once the central bank purchases marketable collateral. Their mechanism is motivated by an increase in the opportunity cost of pledging collateral. Divergent QE effects on financial markets are also supported empirically (see, e.g., Schlepper et al. 2017; Arrata et al. 2020; Corradin and Maddaloni 2020; Roh 2021; Maddaloni and Roh 2021). Koijen et al. (2017) show that in response to the ECB’s purchasing programs, foreign investors sold most of their QE eligible bond holdings to domestic investors pointing to a strong home bias in eligible securities. This shift was also documented in aggregate data by Avdjiev, Everett, and Shin (2019).

¹⁷ Under the umbrella of the PSPP, the ECB started in 2015 to buy nominal and inflation-linked government bonds as well as securities issued by recognized agencies, regional and local governments, international organizations, and multilateral development banks located in the euro area. Overall, around 90% of purchases correspond

**Figure 6****Special collateral repo market**

Panel A depicts the total trading volume in the special collateral market for trades involving eligible and noneligible collateral. Panel B depicts the spread between the average repo rate on noneligible and eligible securities (average rate for noneligible collateral minus average rate for eligible collateral).

retrospectively to compare time trends between (*hypothetically*) *eligible* and *noneligible* assets, which creates a difference-in-difference estimation setting. Observing similar reactions of both types of assets before QE would imply common trends and would allow us to interpret the QE impact as causal.

For this part of the analysis, we focus on the special repo segment that specifies a specific bond as collateral. Figure 6, panel A, shows the trading volume for QE eligible and noneligible securities over time, while panel B depicts the spread between the average repo rate on noneligible and eligible securities. A second, new stylized fact emerges as we observe an increase in special collateral trading volume since the start of QE, an increase that is predominantly driven by QE eligible assets. We also observe that since the start of QE, repo rates for QE eligible assets have fallen below those of noneligible assets in line with Arrata et al. (2020) and Corradin and Maddaloni (2020). The spread between noneligible and eligible assets was always positive during the QE period and has increased further after the expansion and extension of the program. This is in line with the idea that QE eligible assets become scarcer due to central bank asset purchases and the demand to source those assets in the repo market has increasingly led to higher repo specialness. Naturally, one expects collateral types that are scarcer to be more collateral-driven and thus less connected to funding-based money market segments.

to government bonds. The implementation provisions specify the conditions under which the ECB (via local central banks) is allowed to purchase government bonds: they (a) contain a maturity restriction that specifies the minimum and maximum remaining maturity of a security, (b) contain a yield restriction that states that the yield of a security needs to be above the ECB's DFR, and (c) restrict purchases to bonds denominated in euro. Over time, the ECB has adjusted and modified the initial implementation provisions. For example, the yield restriction ceased to exist at the end of 2017.

1.4 Data

For our analysis, we employ high-frequency, trade-level data for the European repo market for the time period between 2010 and 2020. Our data includes all electronically traded repo transactions in euro on the three main trading platforms (i.e., BrokerTec, Eurex, and MTS) and covers more than 70% of the entire repo market universe. For each transaction, we observe the trade date, the term, the trade volume, the rate, the collateral identified by a unique ISIN (for special repos) or basket (for GC repos), the lender, the borrower, the aggressor type and the trading platform.

We exclude a very small fraction of repo contracts and trading days to ensure the robustness and general validity of our results. First, we focus on the repo term types Overnight (ON), Tomorrow-Next (TN), and Spot-Next (SN), with the purchase date being tonight, tomorrow, or the day after tomorrow, respectively, and the repurchase date 1 day thereafter.¹⁸ Second, we exclude holidays as well as year-end trading days. We further provide robustness checks excluding all quarter-end and end of ECB maintenance period trading days, as well as the corresponding weeks.¹⁹ Third, we exclude special repos secured by corporate securities. Fourth, we exclude repos with floating rates, repos with open term type, bilaterally prearranged repos as well as repos that are not cleared via a CCP. Finally, we exclude repos that are traded infrequently.²⁰ We also perform our analyses for three different groups of countries: (a) Germany, (b) core European countries, and (c) all European countries.²¹

To identify banks, we follow the approach of Ranaldo, Schaffner, and Tsatsaronis (2019) and Di Filippo, Ranaldo, and Wrampelmeyer (2022) based on supervisory data. We then classify banks into access and nonaccess institutions depending on whether they need to fulfill the reserve requirements of the Eurosystem and have access to the deposit facility.²² Our data contains repo trades involving 98 different banks, of which 85 are access banks and 13 are nonaccess banks.²³ We observe information on both the lending and

¹⁸ These three term types make up 97% of the entire repo market trading volume. Trading in the GC market predominantly takes place in the ON and TN segments, whereas trading in the special repo market segment predominantly takes place in the TN and SN segments. We account for this in the empirical part.

¹⁹ On quarter- and year-end days as well as on end of ECB maintenance period days, window dressing affects repo rates (Ranaldo, Schaffner, and Vasios 2021).

²⁰ To be included in our analysis, a repo needs to be traded at least 100 times. In addition, between the issuance and maturity of the underlying collateral, a repo needs to be traded at least once every 2 weeks in 95% of the time. Our results are robust to different specifications.

²¹ Core European countries include Austria, Belgium, Finland, France, Germany, and the Netherlands; all European countries include in addition the European Union, Ireland, Italy, Portugal, and Spain.

²² Banks trading in the repo market are, for example, Deutsche Bank AG and Nordea Bank Danmark A/S. The former is a euro area bank with access to the deposit facility, while the latter is a foreign bank without access. For our classification, we assume that local subsidiaries of global banking institutions operate independently in the short run. Thus, euro area subsidiaries of foreign banking groups have access to the deposit facility, while foreign subsidiaries of euro area banking groups do not have access.

²³ The number of nonaccess banks is constant throughout our sample, thereby mitigating endogeneity concerns of nonaccess banks switching their location to access the deposit facility.

Table 1
Breakdown of the repo data

	Access banks	Nonaccess banks
General collateral euro area repos		
Trade size (mn)	206.0	133.7
Repo rate (GC > DFR)	0.16%	0.13%
Repo rate (GC < DFR)	−0.22%	−0.29%
Interquartile range	0.51%	0.66%
# baskets traded in per month	4.0	3.7
Total assets (bn)	290.6	241.2
Leverage ratio	16.7	17.2
	QE eligible assets	QE noneligible assets
Special collateral euro area repos		
Trade size (mn)	22.3	21.6
Repo rate (pre-QE)	0.16%	0.26%
Repo rate (post-QE)	−0.58%	−0.13%
Interquartile range	0.60%	0.74%
Bond issue size (bn)	37.8	36.7
Bond tenor (years)	12.6	6.4
Bond coupon rate	3.2%	3.1%

The table shows the breakdown of the repo data denominated in euro. For the classification between access and nonaccess banks, we focus on the lending side in GC repos; for the classification between eligible and noneligible assets, we focus on special repos. For all repo trades, we show the average daily trade size (in million euro), the volume-weighted, average daily repo rate for periods when the average GC rate is above or below the deposit facility rate (for periods prior to and after the introduction of the PSPP QE program), and the average daily interquartile range defined as the difference between the 25th and 75th percentile repo rate. For access and nonaccess banks, we further show the average number of different GC baskets each bank is trading in during a month as well as the average total asset size (in billion euro) and leverage ratio at the end of our sample period. For eligible and noneligible assets, we show each bond's issue size (in billion euro), tenor (in years), and coupon rate (if bonds are fixed rate).

borrowing banks for trades featuring 59% of the entire trading volume; among those trades, 22% are associated with a nonaccess bank. The upper part of Table 1 provides a comparison of the two groups. We see that the characteristics of access and nonaccess banks are similar. For instance, at the end of our sample period, access banks had, on average, assets worth 290.6 billion euro compared to 241.2 billion euro for nonaccess banks, the leverage ratios were about 17 for both types of banks.

Interestingly, access lenders trade, on average, at a higher rate than nonaccess banks (0.16% compared to 0.13%) in the GC > DFR environment, which reflects, for example, their lower market order share as shown in Figure 5. In the GC < DFR environment, the difference more than doubles as the average repo rate of trades involving access lenders is −0.22% compared to −0.29% for nonaccess lenders. This points to the idea that access lenders move away from funding-driven baskets trading below the DFR and substitute their investments with DFR deposits that yield a higher remuneration.

Moreover, we classify assets as (hypothetically) eligible and noneligible for QE according to the PSPP's implementation provisions. Our data set contains special repo trades involving more than 2,000 different collateral assets (ISINs). Seventy-six percent of our sample involves repo trades collateralized by (hypothetically) eligible assets, 24% collateralized by noneligible assets.

The lower part of Table 1 compares characteristics between eligible and noneligible collateral assets. We observe that repo rates for QE eligible assets have fallen below those of noneligible assets since the start of QE. Other bond characteristics, such as repo trade size, bond issue size, and bond coupon rate, are comparable between the two groups.²⁴ Since we only consider government bonds as collateral, all assets are broadly accepted as collateral and are treated in the same way for regulatory purposes.

2. Empirical Results

The idea of this paper is that a weaker correlation between money market rates is a sign of segmentation that emerges when the role of collateral in repos becomes dominant to the role of funding. To document this mechanism, we begin by looking at the correlation between secured repo rates and the unsecured money market benchmark rate (EONIA) before broadening our view to other money market segments.

2.1 Access/nonaccess banks

We first want to understand whether the access restrictions to the central bank's deposit facility lead to money market segmentation. Access banks always have the option to store funds at the deposit facility, which is particularly attractive in the $GC < DFR$ environment; during those periods, we expect access banks to predominantly trade in the repo market for collateral purposes. Our first testing hypothesis is, therefore, that lending rates of access banks are more collateral-driven and thus less connected to the funding-based EONIA benchmark rate, in particular when repo rates fall below the DFR.

We provide a first graphical intuition of this mechanism in Figure 7 that illustrates the lower sensitivity of access banks to changes in the EONIA rate in the form of impulse response functions. We compute impulse response functions for trades involving access and nonaccess banks separately for periods during which the average GC rate is above the deposit rate (left panel) and below the deposit rate (right panel). The coefficients are derived from a time series regression of current repo rate changes on the concurrent and the 10 preceding EONIA rate changes. The left panel highlights that access and nonaccess banks react similarly during periods when the GC rate is above the deposit rate with the point estimate for access banks being slightly smaller. Even if the deposit facility provides a smaller remuneration than a repo trade, the storage of liquidity at the facility is convenient for banks with access to it, thus explaining the overall lower sensitivity of access banks. However, once GC rates drop below the DFR, the sensitivity of access banks is almost completely muted as they predominately trade for collateral-purposes in the repo market.

²⁴ The average tenor is longer for eligible assets as this is one of the implementation criteria.

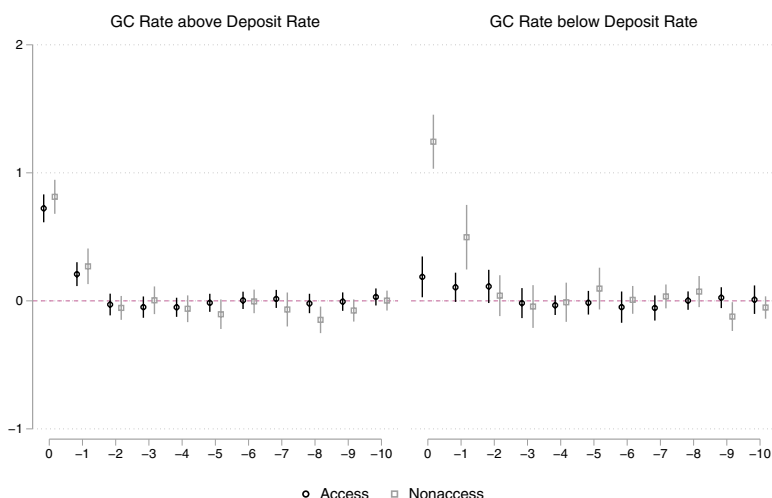


Figure 7
Impulse response for trades involving access/nonaccess banks

The figure depicts the impulse response function of repo rates to changes in the money market benchmark rate for trades involving access and nonaccess banks in the period when the average GC rate is above (left panel) and below the rate on the deposit facility (right panel). The coefficients are derived from a time series regression of current repo rate changes on the concurrent and the 10 preceding EONIA rate changes controlling for month-term fixed effects. The regression includes heteroscedasticity-robust standard errors. Data include GC repo transactions for Germany for the term types ON and TN for the time period 2010–2018.

By contrast, banks without the outside option of storing funds at the deposit facility are still active in funding-based repo market segments keeping them more connected to changes in the EONIA rate.

We formalize the graphical intuition in a set of panel regressions. Our main regression equations read as follows:

$$\Delta r_{t,i,l}^{GC} = \beta_1 \cdot \Delta MMRate_t + \beta_2 \cdot D_{t,n}^{Dep} + \beta_3 \cdot \Delta MMRate_t \cdot D_{t,n}^{Dep} + \beta_4 \cdot \Delta r_{t-1,i,l}^{GC} + \epsilon_t \quad (1)$$

$$\Delta r_{t,i,l}^{GC} = \beta_1 \cdot \Delta MMRate_t + \beta_2 \cdot D_{t,l}^{Access} + \beta_3 \cdot \Delta MMRate_t \cdot D_{t,l}^{Access} + \beta_4 \cdot \Delta r_{t-1,i,l}^{GC} + \epsilon_t \quad (2)$$

$$\Delta r_{t,i,l}^{GC} = \beta_1 \cdot \Delta MMRate_t + \beta_2 \cdot D_{t,n}^{Dep} + \beta_3 \cdot D_{t,l}^{Access} + \beta_4 \cdot \Delta MMRate_t \cdot D_{t,n}^{Dep} + \beta_5 \cdot \Delta MMRate_t \cdot D_{t,l}^{Access} + \beta_6 \cdot \Delta MMRate_t \cdot D_{t,n}^{Dep} \cdot D_{t,l}^{Access} + \beta_7 \cdot \Delta r_{t-1,i,l}^{GC} + \epsilon_t, \quad (3)$$

where $\Delta r_{t,i,l}^{GC}$ denotes the log change in GC repo rates of basket i and lender type (access vs. nonaccess) l at time t and $\Delta MMRate_t$ denotes the log change in the EONIA rate which we refer to as the money market benchmark rate. Moreover, we employ two dummy variables: $D_{t,n}^{Dep}$, which is equal to one if country n 's

average GC rate is below the deposit facility rate, and $D_{t,l}^{Access}$, which is equal to one if the lender l has access to the deposit facility.²⁵ We add month-term fixed effects and employ heteroscedasticity-robust standard errors. Since trading in the GC repo market is concentrated in the ON and TN term types, we show our main results as a pooled regression of both term types in Table 2. We report our results for (a) Germany in columns 1–3, (b) core European countries in columns 4–6, and (c) all countries in columns 7–9.

As a first step, we consider repo transactions collateralized by German government bonds. Since German collateral is considered to be safe and liquid, this limits any concerns about cross-country differences in sovereign risk and liquidity affecting our results. Regression (1) relates changes in GC rates to changes in the money market benchmark rate, depending on whether the GC rate is above or below the DFR. The results highlight that GC rates react strongly: a one-percentage-point increase in the EONIA rate is accompanied by an increase in GC rates of about 61 bps. The effect is smaller at 54 bps when the GC rate is below the DFR. This confirms our intuition that GC repo trades have a strong funding motive, which, however, becomes less important in the $GC < DFR$ environment.

In Regression (2), we analyze the different reactions of access and nonaccess banks. GC rates lent by banks with access to the deposit facility react less strongly. An increase in the funding-based benchmark rate by one percentage point relates to an increase in GC rates involving access banks of 55 bps compared to 76 bps for nonaccess banks.

Considering our main Regression (3), which includes both dummy variables, we observe a combined effect: GC rates involving lenders with access react less, their reaction is particularly weak when GC rates are below the DFR. The effect of changes in the money market benchmark rate on GC rates is 71 bps for nonaccess banks compared to 58 bps for access banks during periods when GC rates are above the DFR. Once GC rates fall below the DFR, the effect is almost one-to-one for nonaccess banks, while it decreases to 28 bps for access banks. This indicates that lenders with access to the deposit facility hardly react to changes in the funding-based money market rate in the $GC < DFR$ environment as their trades are predominantly driven by collateral demand, while lenders without access are very sensitive to it.

Columns 4–9 expand our analysis by looking at larger samples consisting of core European countries and all European countries. Overall, the results are consistent with the effect being strongest in the German “safe haven” market. The reason for it is that repo rates for Germany are the lowest across the euro

²⁵ The denominations are: $\Delta r_{t,i,l}^{GC}$ is the log change in the volume weighted average daily repo rate per basket and lender type in percentage points. Correspondingly, $\Delta MMRate_t$ denotes the log change in the EONIA rate expressed as percentage points.

Table 2
ECB access

	Germany			Core			All		
	(1) $\Delta repo^{GC}$ ON/TN b/t	(2) $\Delta repo^{GC}$ ON/TN b/t	(3) $\Delta repo^{GC}$ ON/TN b/t	(4) $\Delta repo^{GC}$ ON/TN b/t	(5) $\Delta repo^{GC}$ ON/TN b/t	(6) $\Delta repo^{GC}$ ON/TN b/t	(7) $\Delta repo^{GC}$ ON/TN b/t	(8) $\Delta repo^{GC}$ ON/TN b/t	(9) $\Delta repo^{GC}$ ON/TN b/t
$\Delta MMrate$	0.612*** (15.672)	0.762*** (10.955)	0.708*** (8.515)	0.531*** (24.119)	0.741*** (17.871)	0.698*** (15.026)	0.480*** (26.602)	0.641*** (18.235)	0.610*** (16.458)
D^{Dep}	-0.027 (-1.271)	-0.029 (-1.366)	-0.029 (-1.366)	-0.017* (-1.731)		-0.018* (-1.783)	-0.001 (-0.228)		-0.001 (-0.228)
$\Delta MMrate \cdot D^{Dep}$	-0.073 (-0.886)		0.294** (2.274)	0.032 (0.603)		0.318*** (3.976)	0.101** (1.994)		0.418*** (5.472)
D^{Access}		0.002 (0.265)	0.003 (0.308)		-0.002 (-0.346)	-0.001 (-0.220)		-0.004 (-1.023)	-0.003 (-0.919)
$\Delta MMrate \cdot D^{Access}$		-0.216*** (-2.697)	-0.128 (-1.358)		-0.271*** (-5.758)	-0.217*** (-4.144)		-0.209*** (-5.252)	-0.176*** (-4.184)
$\Delta MMrate \cdot D^{Access} \cdot D^{Dep}$			-0.590*** (-3.730)			-0.436*** (-4.328)			-0.461*** (-4.753)
N	10,172	10,172	10,172	35,650	35,650	35,650	59,029	59,029	59,029
R^2	.225	.228	.232	.198	.203	.204	.192	.195	.196
$\Delta repo^{GC}$ lagged	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

The table reports the regression results examining the impact of access to the ECB's deposit facility on money market segmentation. The dependent variable is the change in the GC rate $\Delta repo^{GC}$. $\Delta MMrate$ denotes the change in the EONIA rate. D^{Dep} equals one if a country's average GC rate is below the deposit facility rate. D^{Access} equals one if a lending bank has access to the deposit facility. t -statistics are in parentheses. All regressions include month-term fixed effects and heteroscedasticity-robust standard errors. Data include GC repo transactions for Germany, core European countries and all European countries pooled across the term types ON and TN for the time period 2010–2018. * $p < .1$; ** $p < .05$; *** $p < .01$.

area, hence the effect of access banks selecting themselves into more collateral-driven baskets is strongest in this environment.²⁶

We perform a number of additional robustness checks which confirm our main results. First, we ensure that the results are consistent for each term type. Second, we show that the results are robust for different standard error and fixed effect specifications.²⁷ Third, we repeat our analysis in a sample without quarter end and end of ECB maintenance period days and weeks. Our results are robust to removing those days that are characterized by higher idiosyncratic rate movements and spikes. And finally, we replicate our results for the subperiod until 2016 to account for the lower volatility and trading volume in the EONIA rate in recent years. The Internet Appendix reports all the results.

To gain additional insight into our mechanism and the role of the DFR reflecting the opportunity cost of engaging in a repo trade for funding-purposes, we extend our analysis by looking at the distance between GC rates and the DFR. Our idea is that the sensitivity of access banks to changes in funding-based money market rates is weaker for periods when the opportunity cost of engaging in a repo trade for funding-purposes is higher. More precisely, we expect access banks to be increasingly active in collateral-based repo market segments, which are more disconnected from funding-based money market rates, the lower GC rates are relative to the DFR.

Table 3 reports the regression results which focus on the distance of repo rates to the DFR. We show two regression specifications: (a) by employing a new, continuous variable *DFR distance*, which measures the difference between the DFR and the GC rate;²⁸ and (b) by employing three buckets with *DFR1* indicating periods when the GC rate is 25- to 50-bps above the DFR, *DFR2* indicating periods when the GC rate is 0- to 25-bps above the DFR, and *DFR3* indicating periods when the GC rate is below the DFR. For all regressions, we replace our previous D^{Dep} dummy with the newly introduced DFR distance variable or the respective DFR distance buckets, interacted with the change in the EONIA rate $\Delta MMrate_t$. We report our results for (a) Germany in columns 1 and 2, (b) core European countries in columns 3 and 4, and (c) all countries in columns 5 and 6.

Our results highlight that the distance of GC rates to the DFR significantly affects access banks. Regression (1) shows that the sensitivity of access banks to changes in the money market benchmark rate decreases in the distance between the DFR and the GC rate. For example, if the difference between the

²⁶ We show in the Internet Appendix that the effect is smallest if we only consider periphery euro area countries as (a) repos secured by those assets are predominantly funding-driven and (b) the corresponding repo rates are trading above or close to the DFR.

²⁷ The results are also close to and consistent with our benchmark specification when we account for basket-month-term fixed effects which suggests that the higher funding immediacy of nonaccess banks already manifests itself in the same basket.

²⁸ Positive values of the DFR distance variable indicating that GC rates are below the DFR.

Table 3
ECB access: Distance-to-deposit facility rate

	Germany		Core		All	
	(1) $\Delta repo^{GC}$ ON/TN b/t	(2) $\Delta repo^{GC}$ ON/TN b/t	(3) $\Delta repo^{GC}$ ON/TN b/t	(4) $\Delta repo^{GC}$ ON/TN b/t	(5) $\Delta repo^{GC}$ ON/TN b/t	(6) $\Delta repo^{GC}$ ON/TN b/t
$\Delta MMrate$	0.762*** (10.951)	0.762*** (10.945)	0.742*** (17.878)	0.742*** (17.887)	0.641*** (18.239)	0.642*** (18.248)
D^{Access}	0.001 (0.103)	0.001 (0.092)	-0.003 (-0.639)	-0.004 (-0.791)	-0.005 (-1.321)	-0.006 (-1.560)
$\Delta MMrate \cdot D^{Access}$	-0.266*** (-2.978)	0.067 (0.742)	-0.343*** (-6.661)	0.103* (1.813)	-0.261*** (-6.022)	0.124** (2.424)
$\Delta MMrate \cdot D^{Access} \cdot DFRDistance$	-0.324** (-2.133)		-0.468*** (-5.095)		-0.294*** (-3.665)	
$\Delta MMrate \cdot D^{Access} \cdot DDFR1$		-0.224** (-2.248)		-0.147** (-2.311)		-0.251*** (-3.547)
$\Delta MMrate \cdot D^{Access} \cdot DDFR2$		-0.276*** (-3.495)		-0.419*** (-8.699)		-0.371*** (-8.459)
$\Delta MMrate \cdot D^{Access} \cdot DDFR3$		-0.548*** (-5.355)		-0.494*** (-7.230)		-0.387*** (-5.805)
N	10,172	10,172	35,650	35,650	59,029	59,029
R^2	.229	.232	.204	.207	.196	.197
$\Delta repo^{GC}$ lagged	Yes	Yes	Yes	Yes	Yes	Yes

The table reports the regression results examining the impact of access to the ECB's deposit facility on money market segmentation with a focus on the distance of repo rates to the DFR. The dependent variable is the change in the GC rate $\Delta repo^{GC}$. $\Delta MMrate$ denotes the change in the EONIA rate. D^{Access} equals one if a lending bank has access to the deposit facility. $DFRDistance$ refers to the difference between the DFR and a country's average GC rate. $DDFR1$ indicates time periods when the GC rate is between 25- and 50-bps above the DFR, $DDFR2$ indicates time periods when the GC rate is between 0- and 25-bp above the DFR, and $DDFR3$ indicates time periods when the GC rate is below the DFR. t -statistics are in parentheses. All regressions include month-term fixed effects and heteroscedasticity-robust standard errors. Data include GC repo transactions for Germany, core European countries and all European countries pooled across the term types ON and TN for the time period 2010–2018. * $p < .1$; ** $p < .05$; *** $p < .01$.

DFR and the GC rate increases by 10 bps (e.g., GC rates drop by 10 bps relative to the DFR), the sensitivity of access banks to changes in the funding-based money market rate reduces by 3 bps. Regression (2) confirms that access banks already become less reactive to changes in the EONIA when GC rates get closer to the DFR. This effect is intensified once GC rates drop below the DFR. Both results confirm that access banks select themselves into more collateral-driven repo market segments when the opportunity cost of engaging in funding-based repo trades are higher, which leads to a disconnect of their repo lending rates from funding-based money market rates.

2.2 Introduction of ECB tiering as natural experiment

The introduction of the ECB's two-tier system for remunerating excess reserve holdings in 2019 is a natural experiment to illustrate our mechanism. The tiering system provides each access bank with an allowance to store excess reserves at the ECB without paying the negative deposit rate that would

Table 4
ECB access: Introduction of ECB tiering system

	Germany	Core	All
	(1) $\Delta repo^{GC}$ ON/TN b/t	(2) $\Delta repo^{GC}$ ON/TN b/t	(3) $\Delta repo^{GC}$ ON/TN b/t
$\Delta MMrate$	0.901*** (17.522)	0.765*** (13.081)	0.791*** (17.834)
$D^{Tiering}$	-0.016* (-1.822)	-0.021*** (-3.181)	-0.041*** (-4.330)
$\Delta MMrate \cdot D^{Tiering}$	0.145 (0.740)	0.068 (0.342)	0.220 (0.509)
D^{Access}	-0.001 (-0.267)	-0.003 (-1.294)	-0.002 (-0.460)
$\Delta MMrate \cdot D^{Access}$	-0.070 (-0.588)	0.063 (0.681)	0.035 (0.477)
$\Delta MMrate \cdot D^{Access} \cdot D^{Tiering}$	-0.210 (-0.819)	-0.566** (-2.215)	-0.944* (-1.648)
N	503	1,614	2,735
R^2	.464	.366	.339
$\Delta repo^{GC}$ lagged	Yes	Yes	Yes

The table reports the regression results examining the impact of the ECB's introduction of the tiering program on money market segmentation. The dependent variable is the change in the GC rate $\Delta repo^{GC}$. $\Delta MMrate$ denotes the change in the money market rate. $D^{Tiering}$ equals one during the period the tiering was in place beginning on October 30, 2019. D^{Access} equals one if a lending bank has access to the deposit facility. t -statistics are in parentheses. All regressions include month and term fixed effects and heteroscedasticity-robust standard errors. Data include GC repo transactions for Germany, core European countries and all European countries pooled across the term types ON and TN for the time period of the two ECB maintenance periods preceding the ECB tiering introduction as well as the two ECB maintenance periods thereafter (August 2019 to January 2020). * $p < .1$; ** $p < .05$; *** $p < .01$.

generally apply to them.²⁹ The introduction of the tiering implies that the deposit facility becomes an even more attractive option for access banks to store liquidity, in particular for those access banks that still have unused allowances. Fuster, Schelling, and Towbin (2021) find that an increase in exemption allowances in Switzerland induced banks to deposit more funds at the deposit facility and to restrict their lending activities. In the context of our mechanism, access banks have an additional incentive to store funds at the ECB's deposit facility due to the tiering exceptions. Consequently, we expect access banks to move out of funding-based repo market segments after the tiering exemptions were introduced, while continuing to trade in collateral-driven parts of the repo market which are less connected to funding-based money market rates, thereby exacerbating money market segmentation.

To isolate the impact of the tiering system, we adapt our empirical analysis for the time period around the introduction of the two-tier system covering the two ECB maintenance periods before and after the introduction.³⁰ For this, we introduce the dummy variable $D^{Tiering}$, which is equal to one after the

²⁹ For holdings above the allowance, the DFR continues to apply.

³⁰ The tiering is an out-of-sample analysis, and hence it does not affect our main results.

introduction of the tiering system on October 30th, 2019. For all regressions, we replace our previous D^{Dep} dummy with the newly introduced $D^{Tiering}$ dummy, interacted with changes in the EONIA rate.

The results in Table 4 show that access banks are less responsive to EONIA rate changes after the introduction of the tiering. This effect is significant for core and all European countries in line with the idea that banks in peripheral countries have benefited most from the introduction of the tiering and thus became less reactive to changes in funding-based money market rates. The two-tier system has a strong impact on banks that still have room in their allowance, which is more common in the periphery (Cœuré 2018). Because of the home bias observed among European financial institutions (Kojien et al. 2017), this implies that the segmentation becomes particularly noticeable in GC baskets associated with those countries.

2.3 Eligible/noneligible assets

The eligibility criteria of the QE program are a second illustration of our mechanism. Eligible collateral is scarce and in high demand; repos secured by those scarce assets are more collateral-driven and thus less connected to funding-based money market rates. Our second testing hypothesis is, therefore, that repo rates secured by QE eligible assets are more disconnected from other money market rates, such as the EONIA. Similar to the previous analysis, a lower correlation implies a stronger segmentation between the secured and unsecured money market segments. Our main panel regression equations read as follows:

$$\begin{aligned}\Delta r_{t,i,l}^{Special} = & \beta_1 \cdot \Delta MMRate_t + \beta_2 \cdot D_t^{QE} + \beta_3 \cdot \Delta MMRate_t \cdot D_t^{QE} \\ & + \beta_4 \cdot \Delta r_{t-1,i,l}^{Special} + \epsilon_t\end{aligned}\quad (4)$$

$$\begin{aligned}\Delta r_{t,i,l}^{Special} = & \beta_1 \cdot \Delta MMRate_t + \beta_2 \cdot D_{t,i}^{Eligible} + \beta_3 \cdot \Delta MMRate_t \cdot D_{t,i}^{Eligible} \\ & + \beta_4 \cdot \Delta MMRate_t \cdot D_t^{QE} \cdot D_{t,i}^{Eligible} + \beta_5 \cdot \Delta r_{t-1,i,l}^{Special} + \epsilon_t\end{aligned}\quad (5)$$

$$\begin{aligned}\Delta r_{t,i,l}^{Special} = & \beta_1 \cdot \Delta MMRate_t + \beta_2 \cdot D_t^{QE} + \beta_3 \cdot D_{t,i}^{Eligible} + \beta_4 \cdot \Delta MMRate_t \cdot D_t^{QE} \\ & + \beta_5 \cdot \Delta MMRate_t \cdot D_{t,i}^{Eligible} + \beta_6 \cdot \Delta MMRate_t \cdot D_t^{QE} \cdot D_{t,i}^{Eligible} \\ & + \beta_7 \cdot \Delta r_{t-1,i,l}^{Special} + \epsilon_t,\end{aligned}\quad (6)$$

where $\Delta r_{t,i,l}^{Special}$ denotes the log change in special repo rates, and $\Delta MMRate_t$ denotes the log change in the EONIA rate. Moreover, we employ two dummy variables: $D_{t,i}^{Eligible}$, which is equal to one if security i is (hypothetically) eligible for purchase under the PSPP, and D_t^{QE} , which is equal to one after the

introduction of the PSPP in March 2015. Additionally, we add ISIN-month-term fixed effects and heteroscedasticity-robust standard errors.³¹ Trading in the special repo market is concentrated in the TN and SN term types, we therefore show our main results as a pooled regression of both term types in Table 5. We report our results for (a) Germany in columns 1–3, (b) core European countries in columns 4–6, and (c) all countries in columns 7–9.

Regression (1) relates changes in special repo rates to changes in the money market benchmark rate in the period prior to and after the introduction of the QE program. An increase in the unsecured benchmark rate by one percentage point translates into an increase of around 11 bps in special repo rates in the period prior to the PSPP. During the period of unconventional monetary policy, the effect has been muted. Overall, special repo rates are more collateral-driven and thus react less strongly to changes in the EONIA rate than more funding-driven GC rates, which confirms our intuition. Still, also a special repo trade involves a funding motive and reacts to changes in funding conditions.

In Regression (2), we consider the impact of market segmentation along the lines of asset eligibility for QE in a difference-in-differences setting. The dummy variable $D_{t,i}^{Eligible}$ measures whether the underlying collateral asset fulfills the QE eligibility criteria since the start of the program and whether it had (hypothetically) fulfilled the criteria in the prior period. We observe that trades involving hypothetically eligible assets do not exhibit significantly different changes in repo rates prior to QE; eligible and noneligible collateral assets also respond similarly to changes in the benchmark rate during that period. In the pre-QE period, the common trend assumption, therefore, holds. However, since the start of QE, repo trades involving eligible assets have a 17-basis-points lower sensitivity to changes in the EONIA rate compared to noneligible assets. This speaks to an effect caused by unconventional monetary policy.

Our main Regression (3) captures both effects. The impact of changes in the money market benchmark rate on special repo rates is almost muted during QE, which is in particular driven by trades involving QE eligible assets. In the period after the QE introduction, an increase in the benchmark rate by one percentage point is associated with a 5-bps lower reaction of QE eligible assets. While the overall size of this effect seems small, it represents a 50% reduction relative to the overall sensitivity of special repo rates to changes in the unsecured money market rate. It highlights that assets become even more collateral-driven and, therefore, disconnected from overall money market movements when they are targeted by QE leading to their scarcity.

Columns 4–9 extend our analysis to core and all European countries, respectively, and confirm our results. The effect is again strongest for German collateral assets as those assets tend to have the highest collateral demand.

³¹ The fixed effects capture all bond-specific properties that are constant within a month, for example, issue size or on-the-run status.

Table 5
Asset eligibility

	Germany			Core			All		
	(1) $\Delta repo^{Special}$ TN/SN b/t	(2) $\Delta repo^{Special}$ TN/SN b/t	(3) $\Delta repo^{Special}$ TN/SN b/t	(4) $\Delta repo^{Special}$ TN/SN b/t	(5) $\Delta repo^{Special}$ TN/SN b/t	(6) $\Delta repo^{Special}$ TN/SN b/t	(7) $\Delta repo^{Special}$ TN/SN b/t	(8) $\Delta repo^{Special}$ TN/SN b/t	(9) $\Delta repo^{Special}$ TN/SN b/t
$\Delta MMRate$	0.106*** (19.746)	0.098*** (13.001)	0.109*** (13.196)	0.105*** (31.250)	0.095*** (17.688)	0.103*** (17.850)	0.099*** (30.217)	0.094*** (18.379)	0.101*** (18.370)
DQE	-0.016 (-1.448)	-0.016 (-1.420)	-0.016 (-1.420)	-0.008 (-1.176)	-0.008 (-1.176)	-0.008 (-1.157)	-0.020** (-2.001)	-0.020** (-1.996)	-0.020** (-1.996)
$\Delta MMRate \cdot DQE$	-0.150*** (-15.860)	-0.120*** (-8.160)	-0.120*** (-8.160)	-0.126*** (-19.822)	-0.126*** (-19.822)	-0.106*** (-9.820)	-0.108*** (-17.347)	-0.108*** (-17.347)	-0.091*** (-8.349)
$D^{Eligible}$		0.004 (0.443)	0.004 (0.428)		0.003 (0.580)	0.003 (0.578)		0.001 (0.236)	0.001 (0.214)
$\Delta MMRate \cdot D^{Eligible}$		0.006 (0.537)	-0.005 (-0.462)		0.011 (1.622)	0.002 (0.299)		0.004 (0.594)	-0.004 (-0.556)
$\Delta MMRate \cdot D^{Eligible} \cdot DQE$		-0.172*** (-14.030)	-0.052*** (-2.736)		-0.136*** (-17.421)	-0.030** (-2.211)		-0.116*** (-15.232)	-0.026* (-1.936)
N	301,766	301,766	301,766	706,015	706,015	706,015	943,926	943,926	943,926
R^2	.116	.116	.116	.113	.113	.113	.117	.117	.117
$\Delta repo^{Special}$ lagged	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

The table reports the regression results examining the impact of asset eligibility for quantitative easing programs on money market segmentation. The dependent variable is the change in the special rate $\Delta repo^{Special}$. $\Delta MMRate$ denotes the change in the EONIA rate. DQE equals one during the PSPP. $D^{Eligible}$ equals one if a security is (hypothetically) eligible for purchase under the PSPP. t -statistics are in parentheses. All regressions include ISIN-month-term fixed effects and heteroscedasticity-robust standard errors. Data include special repo transactions for Germany, core European countries and all European countries pooled across the term types ON and TN for the time period 2010–2018. * $p < .1$; ** $p < .05$; *** $p < .01$.

Similar to the previous analyses, we perform a number of additional tests (results are reported in the Internet Appendix). First, to manifest the economic determinants of our mechanism, we extend our analysis by looking at asset scarcity associated with unconventional monetary policy in more detail. Asset scarcity is stronger for those assets which have been QE eligible for a longer period. In line with this intuition, we show that repos secured by assets which have been eligible for purchase by the ECB for a longer period become more collateral-driven and their rates are less connected to the funding-based money market segment. And second, we perform the same comprehensive set of robustness checks, which confirm that our results are statistically and economically consistent.

2.4 Additional analyses

To underline the relevance of our results, we carry out two additional analyses (results are also reported in the Internet Appendix). To analyze the *internal* relevance of our results, we look at the joint impact of banks' access to the central bank's deposit facility and collateral assets' eligibility for QE on money market segmentation.³² We find that both effects lead to rate dispersion and add to one another pointing toward a pervasive money market segmentation. In addition, to highlight the *external* relevance of our results, we show that segmentation emerging from the dominant role of repo collateral (as opposed to funding) is present across different money market segments and also *within* the secured market.³³

3. Implications for Monetary Policy

Our results are relevant for the interpretation of different monetary policies. The introduction of the tiering system is one example that we have analyzed in this paper. Because of the tiering exceptions, access banks have an additional incentive to store funds at the deposit facility and thus predominantly trade

³² To examine the impact of QE on GC repo rates, we compute the share of securities eligible for QE within the pool of collateral assets potentially deliverable into a GC basket as an indicator for the likelihood of obtaining a QE eligible asset as collateral in a GC transaction. In addition, Ballensiefen (2022) shows that certain assets are more likely to be delivered as collateral than others. In particular, it is possible to identify the "cheapest-to-deliver" collateral asset, which is the asset that features the smallest repo specialness.

³³ For this, we experiment with alternative rates to capture funding-based money market conditions. We employ the short-term interest rate benchmark (EONIA) as (a) our baseline rate. In 2017, the ECB announced that the euro short-term rate (€STR) will replace the EONIA as the new short-term benchmark in the euro area. Historical €STR rates date back to March 13th, 2017. As a (b) second rate, we, therefore, consider an EONIA-€STR combination with the €STR rate replacing the EONIA rate after its publication. As a (c) third, unsecured reference rate, we consider the overnight euro LIBOR rate. We also consider a set of derivatives-based, forward-looking overnight interest rates. We employ (d) the overnight point of the Overnight Index Swap (OIS)-implied zero curve, which uses 1-, 3-, and 6-month OIS derivatives, as well as (e) the overnight point of the EURIBOR-implied zero curve, which uses 1-, 3-, and 6-month EURIBOR derivatives. We also consider (f) the 1-week OIS rate. Finally, we employ the (g) rate on the GC Pooling basket which is the primary GC repo funding basket. Any segmentation emerging from the dominant role of repo collateral also should be present in relation to funding-based repo rates.

in collateral-driven parts of the repo market which are less connected to funding-based money market rates. The observed effects are not a failure of the tiering policy, which had the goal of supporting “the transmission of monetary policy through banks to firms and households by lowering banks’ costs caused by negative interest rates” (Bank of Finland 2019), but at the same time, it is important to understand those patterns in money markets which help us evaluate the overall effects of such policies. For example, the excessive usage of the deposit facility by access banks raises concerns about discouraging interbank trading which inhibits price determination (Keister, Martin, and McAndrews 2008).

A similar illustration of our mechanism is if foreign banks had access to the deposit facility, such as in Switzerland (Kraenzlin and Nellen 2015). Amendments to the CRR have also encouraged (access) banks to hold reserves at the central bank. While this is beneficial from a financial stability perspective, any evaluation of this policy should take other effects, such as a resultant money market segmentation, into account.

Focusing on QE, the notion that unconventional policies “safeguard the transmission of our monetary policy,” as pointed out by ECB President Christine Lagarde (European Central Bank 2020), should also consider that those programs can create rate dispersion, thereby leading to unintended consequences, such as segmentation across different money market segments.

The money market friction which we document in this paper is also informative for the recent discussion about new short-term benchmark rates as it underlines that measuring the price of cash becomes increasingly difficult in a collateral-driven repo market environment. Consequently, the ECB decided to choose an unsecured rate as their new benchmark.

Our results also have *suggestive* implications for the monetary policy transmission process. The “monetary policy transmission pipeline” involves three distinct steps: (1) the central bank’s monetary policy actions pass-through into the money market, (2) the conditions in the money market impact debt and equity markets, and (3) monetary policy changes then propagate into the real sector.³⁴ Focusing on the first step, “in an idealized money market, any change in the main monetary policy rate should pass through perfectly to all money market rates” (Corradin et al. 2020, p. 13). Our empirical analysis of money market segmentation, by contrast, indicates a dispersion in short-term rates, which would suggest less control of the monetary policy transmission for central banks and pass-through inefficiencies.

To graphically document this effect, we calculate one prime indicator for pass-through inefficiency in money markets following Duffie and Krishnamurthy (2016, p.36): a volume-weighted absolute dispersion index. We present

³⁴ We thank our discussant Olivier Wang for suggesting this metaphor. While the main focus of this paper is on the money market, Wang (2018), among other recent papers, consider the longer-run impacts of an inhibited monetary policy transmission, such as the impact on inflation.

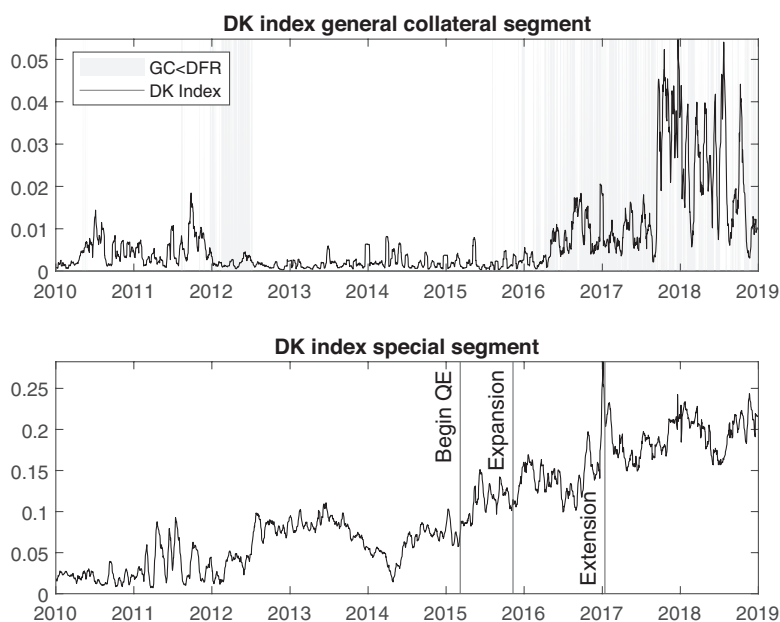


Figure 8
Repo market dispersion

The figure depicts two DK dispersion indexes defined as the volume-weighted average of the absolute deviation of repo rates from the volume-weighted mean repo rate in the spirit of Duffie and Krishnamurthy (2016, p. 36). For the first DK index in the special market, we differentiate between rates on repos secured by QE eligible and noneligible collateral. For the GC segment, we consider rates of trades involving access and nonaccess lenders.

dispersion measures for the GC and special repo market segments in Figure 8, which we accordingly refer to as “DK index.” We observe that the dispersion in the GC segment increases in the GC < DFR environment (as indicated by the grey shaded area). Similarly, we observe an increase in the dispersion in the special segment since the introduction of QE that has further increased with extensions and expansions of the QE program. Both indexes point toward rate dispersion and thus a potential weakening in the monetary policy transmission associated with the two disconnecting mechanisms which we analyze, that is, access to central bank facilities and QE eligibility. This raises the question how the second and third step of the monetary policy pass-through pipeline are affected by the dispersion in money market rates that arises when the role of collateral in repos becomes dominant over the role of funding.

To empirically analyze this idea, we examine the pass-through of changes in the EONIA rate into lending rates faced by corporate borrowers and private households, depending on the conditions in the repo market. The dependent variable is the change of a given lending rate Δr^L , for which we consider borrowing costs of nonfinancial firms and loans to households for house purchases. $\Delta MMRate_t$ denotes the change in the EONIA rate. D^{DKGC} equals

Table 6
Repo dispersion and the pass-through to lending rates

	(1) Nonfin. corporate Δr^L b/t	(2) Nonfin. corporate Δr^L b/t	(3) New housing Δr^L b/t	(4) New housing Δr^L b/t
$\Delta MMrate$	0.506*** (3.431)	0.501*** (3.327)	0.787** (2.804)	0.792** (2.778)
$\Delta MMrate \cdot D^{DKGC}$	-0.522*** (-3.254)		-0.690** (-2.326)	
$\Delta MMrate \cdot D^{DKspecial}$	-0.445 (-1.653)		-0.570*** (-3.318)	
$\Delta MMrate \cdot D^{DKrepo}$		-0.526** (-3.072)		-0.723** (-2.286)
N	1,101	1,101	1,017	1,017
R^2	.126	.125	.174	.173

The table reports the regression results examining the pass-through of changes in the EONIA rate into lending rates faced by corporate borrowers and private households. The dependent variable is the change of a given lending rate Δr^L . Nonfinancial corporate borrowing rates refer to the annualized borrowing costs of nonfinancial firms for new loans, while new housing rates refer to bank interest rates on new loans to households for house purchases with an initial rate fixation period of between one and five years. Both lending rates are available from the ECB's monetary financial institutions (MFI) interest rate statistics. $\Delta MMrate$ denotes the change in the EONIA rate. D^{DKGC} equals one if the dispersion measure for the GC market is above its median. $D^{DKspecial}$ equals one if the dispersion measure for the Special market is above its median. D^{DKrepo} equals one if any of the GC and special repo market dispersion measures is above its median. t -statistics are in parentheses. All regressions include country-year fixed effects and standard errors accounting for clustering at the year level. Data are at a monthly frequency for all European countries for the time period 2010–2020. * $p < .1$; ** $p < .05$; *** $p < .01$.

one if the dispersion measure for the GC market is above its median, $D^{DKspecial}$ equals one if the dispersion measure for the special market is above its median, and dummy D^{DKrepo} equals one if at least one of the two (GC and special) repo market dispersion measures is one.

Table 6 reports the results of our panel regressions. Regression (1) relates changes in nonfinancial corporate borrowing rates to changes in the money market benchmark rate, depending on the dispersion in GC and special repo rates; regression (2) considers our repo market dispersion dummy. The results highlight that lending rates react strongly to changes in money market funding conditions: a one-percentage-point increase in the EONIA rate is accompanied by an increase in corporate borrowing rates of about 50 bps. The effect is, however, almost muted when the dispersion in repo rates is high. Regressions (3) and (4) confirm our results for residential housing rates.³⁵

Both results provide suggestive support for Duffie and Krishnamurthy (2016), who highlight that a dispersion across money market interest rates is a primary indicator of the inefficiency of monetary policy pass-through. Although the monetary policy transmission into the real economy involves

³⁵ In all regressions, we account for country-year fixed effects. We also show in the Internet Appendix that our results remain statistically and economically consistent if we consider other money market rates and if we shorten the sample period to the end of 2016.

additional steps that deserve a detailed analysis beyond short-term rates, our results clearly speak to the importance of the repo market for the monetary policy transmission. Since the repo market is the predominant source of short-term funding, the repo market determines bank funding conditions and ultimately seems to affect the transmission of monetary policy into the real sector. Our results suggest that the money market is more segmented in a collateral-dominant environment and that the consequent rate dispersion is associated with a reduced responsiveness of real-sector lending rates to changes in EONIA rate, which serves as the operational target for monetary policy.

4. Conclusion

As the ECB highlights in its consideration to develop a new short-term interest rate benchmark, “the repo market has a number of unique characteristics related to the motivation for entering into a trade” (European Central Bank 2022). Our analysis has shown that when the motivation to obtain collateral rather than cash is prevalent, the money market becomes more segmented. We uncover two important sources for this segmentation: whether the money market participant is a bank that has access to the central bank’s deposit facility and whether the collateral asset is a target of the QE program. We demonstrate that the lending rates of banks with access to the central bank’s deposit facility and repo rates secured by assets eligible for QE programs are more collateral-driven and thus disconnected from funding-based benchmark rates.

Since the EONIA is the operational target for the ECB, our findings provide suggestive evidence that money market segmentation weakens the monetary policy transmission. Our results also underline that measuring the price of cash becomes increasingly difficult in a collateral-driven repo market environment. It is, therefore, important to consider the mechanism brought to light in this study when evaluating different monetary policies, which is also relevant for the U.S. market, among other regions.

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